PURDUE UNIVERSITY | SCHOOL OF ENGINEERING TECHNOLOGY

MFET 163: Foundations in Geometric Modeling & Data Management

Syllabus

SPRING 2026 SEMESTER

Instructor: Travis Fuerst, CM2-C, PMP, Assistant Professor of Engineering Technology
Purdue Profile:
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Office:
Office phone:
Office hours:
Email:

***Note**: This is the best method of communication.*
Will respond to emails within 24 hours, but may be delayed during travel (see course announcements for further information)

Teaching Assistants & Resources

Course Website	
Course ChatBot	
Course Assistance Request Form	
WL Office Hrs (M-Th 5:30p-7:30p Eastern) Prof Fuerst Office Hrs (By Appointment)	

Course Information

Course number and title: MFET 163 Foundations in Geometric Modeling & Data Management

Meeting day(s) and time(s): |Lecture |WL |
|Lecture |WL |
Labs |

Instructional Modality: Synchronous (Lecture/Lab)

Course credit hours: 2

Prerequisites: None

1 Course Description

This course introduces the process of 3D geometric modeling, and the construction techniques used in the creation of constraint-based solid and surface models. Part modeling and assembly modeling are included, as well as manipulation of the geometric model. Emphasis is on the use of the design process as a problem-solving method and the capture of design intent and modeling behavior to enable the downstream use of 3D solid and surface modeling databases. The role of the 3D model in the overall product design process and the place of the geometric product definition in the product lifecycle are covered. Product Data Management (PDM) and Product Lifecycle Management (PLM) are covered as they relate to the design process. 2 credit hours.

2 Learning Objectives

2.1 Explain the Product Design Process.

- 2.1.1 Distinguish between the different product design processes.
- 2.1.2 Explain the most appropriate design process for a given project.
- 2.1.3 Leverage project management techniques to manage individual and group work.
- 2.1.4 Explain the importance of freehand sketching as a communication tool for the designer.
- 2.1.5 Construct clear freehand Multiview and pictorial freehand sketches to convey design ideas.

2.2 Explain the Role of Parametric Solid Modeling in the Product Lifecycle.

- 2.2.1 Summarize the history of Engineering Graphics.
- 2.2.2 Illustrate the future of geometric modeling in the product lifecycle.
- 2.2.3 Work within a Product Data Management (PDM) system to support design, collaboration, and geometry re-use to simulate an industry PLM setting.

2.3 Demonstrate proficiency in the use of high-end Computer-Aided Design (CAD) applications.

- 2.3.1 Effectively utilize a CAD system.
- 2.3.2 Create 3D geometry from 2D sketched geometry.
- 2.3.3 Construct features using advanced modeling techniques such as Patterns and Transformations.
- 2.3.4 Interrogate a simple Technical Data Package to interpret GD&T in the form of Model-Based Definition (MBD)
- 2.3.5 Construct and simulate digital mock-up visualizations and kinematic simulations.
- 2.3.6 Demonstrate the ability to work across CAD systems to build a product.

2.4 Comprehend design intent in the design of a product.

- 2.4.1 Analyze a part/product based on its use, context, and variants to build in appropriate design intent.
- 2.4.2 Create models using parameters to create variants and populate part libraries.
- 2.4.3 Develop product structures that allow for product variation and re-use.

2.5 Describe Product Data Management and its role in Parametric Modeling.

- 2.5.1 Describe the concept of Product Data Management.
- 2.5.2 Explain the difference between PDM and PLM.
- 2.5.3 List the primary functions of PDM.

2.6 Follow instructions using listening, reading, and analytical skills.

- 2.6.1 Effectively follow instructions in a working/learning environment.
- 2.6.2 Utilize instructions to complete this course successfully.
- 2.6.3 Decipher confusing instructions and know how/when to ask for clarification.
- 2.6.4 Stay organized to be able to find what you need when you need it.

3 Course Website

Brightspace is our course management system. You can access the course website at purdue.brightspace.com. It is strongly suggested that you explore and become familiar not only with the site navigation but also with the content and resources available for this course. For example, you will find on the left-hand course menu the items Handy References, Student Help, and Student Success. Handy References provides links to resources used in this course. Student Help contains information about Brightspace, including tutorials to help you learn about Brightspace functionality, as well as a link to the Purdue Libraries. Student Success contains many resources offered through programs at Purdue to help you become successful in your courses, such as tutoring, downloadable resources, and more.

Additionally, a course information site has been set up to make key information more accessible, such as the course syllabus, due date schedule, office hour information, and how-to videos at...

4 Course Structure & Methodology

4.1 This section exists to provide insight to the student on how and why the course is set up and administered the way it is. Just like the company you are striving to lend your talents to, this course has standards and processes. If you do not meet those standards and adhere to those processes, you will not be compensated well, if at all. Therefore, the course is structured to mimic the workings of a design engineering organization:

4.1.1 Course Instructor - fills the role of the ***Senior Design Leader*** and is responsible for defining work to be completed as well as setting, instructing, and enforcing the organization's processes and standards.

4.1.2 Course Teaching Assistant - fills the role of ***First Line Leaders*** and is responsible for students in their assigned sections. In some cases, TAs will enforce course standards by reviewing student work and awarding the appropriate number of points based on the quality of the work and its timely submission.

4.1.3 Graders – fill the role of Quality Inspector and assist TAs by reviewing student work against the designated rubric to evaluate and assign points to completed assignments based on adherence to course standards and given rubric.

4.1.4 Students - fill the role of an ***Independent Contractor,*** compensated for the quality of their assigned work package (assignment, project, quiz, exam) delivered on time. If the work package does not meet the prescribed standards, it will be compensated accordingly. All work packages must be submitted using the proper process, system, and prescribed deadline to be considered for points compensation. Students must meet all requirements set forth for each work package to receive full credit. It is the student's responsibility to purchase all required course materials. In the industry, if your work is not up to standard or incomplete, you may not be fully paid or may not receive opportunities for future work.

4.1.5 Course Syllabus – As a student in the course, fulfilling the requirements of an ***Independent Contractor***, you are not a salaried employee. All compensation is awarded on a piecework basis, meaning you must produce a project to standard to be fully compensated. You are not automatically guaranteed or entitled to compensation simply because you are enrolled in the course. The course syllabus is the contract between the Student/Independent Contractor and the Course Instructor/Senior Design Leader.

4.1.6 Intellectual Property – all requirements, presentations, lectures, quizzes, exams, and any other materials supplied by the instructor are the copyrighted intellectual property of the ***Course Instructor/Senior Design Leader*** or a third-party organization. All work produced by S***tudents/Independent Contractors*** is considered their individual intellectual property and must always be tightly secured by the ***Student/Independent Contractor***. Any instance of a ***Student/Independent Contractor's*** intellectual property being used by another Student/Independent Contractor in this course or a similar course, or any future cases of this course, will result in an Academic Integrity or Copyright investigation with appropriate consequences. Refer to section 24 – Academic Integrity.

4.1.7 Technology – the course will leverage several technologies, both academic and professional. Brightspace will serve as a communication medium between the Course Instructor/Senior Design Lead and the Teaching Assistant/First Line Leader, a delivery tool for work package requirements, and a platform for recording grades/compensation. Teamcenter will be used to store and submit all CAD and hand sketching-related work packages for grading/compensation. NX will be used to create all CAD-related work packages.

4.1.8 Business Processes – to be fully compensated for work packages, all prescribed processes must be adhered to.

4.1.9 Attendance – the ***Course Instructor/Senior Design Lead*** is required to be present and prepared for all Lecture unless it is deemed not necessary for a given week or due to unforeseen circumstances. ***Teaching Assistants/First Line Leaders*** are required to be prepared for and present in all assigned Labs. Suppose unforeseen circumstances prevent them from being in the assigned lab. In that case, they must coordinate with other Teaching Assistants/First Line Leaders before their absence to ensure coverage of the missed lab and notify the ***Course Instructor/Senior Design Leader***. ***Students/Independent Contractors*** should be present at all lectures to receive the requirements and standards for the week's work packages. They must be present in their assigned lab section to complete all assignments or remain in the lab section until the end of the period to be counted as present for attendance. If ***Students/Independent Contractors*** do not attend the prescribed lab section, they will not be compensated for that session. Within industry, failing to notify one's employer of absence without a valid reason does not create a positive impression, suggests a lack of accountability for completing work, and will be considered during the annual contract review, particularly when determining eligibility for salary increases or promotions.

5 Required Texts

5.1 This course uses a required digital textbook developed to support modern CAD, PDM, and Design for Change workflows as they are practiced in industry. The resource was created to provide an integrated, up-to-date reference that no single traditional textbook adequately covers. All labs, assignments, and assessments are aligned with this material, and students are expected to use it as a technical reference throughout the course. It is intended to support future coursework, internships, and professional practice. As with any engineering resource, the value derived from it depends on the extent to which it is engaged; course expectations and assessment standards remain the same regardless of individual usage preferences.

5.2 From CAD to PDM: Core Concepts for Industry 4.0

Link to purchase: From CAD to PDM: Core Concepts for Industry 4.0

(provides 4 years of access to the content)

5.2.1 If you have any trouble with your purchase of the textbook, you can find solutions at Customer Service - eAcademic Books.

5.2.2 If you have previously taken this course and purchased previous textbooks, you can email the publisher and they will waive the purchase price of this textbook based on your last purchase records.

5.2.3 This course has significant reading requirements from the above text for knowledge that will be needed for modeling, quizzes, and exams. All quizzes will be taken using the publisher's website.



6 Minimum Technology/Equipment Requirements

6.1 Headphones or earbuds should be brought to the lab so that you can watch any lecture or tutorial content without disturbing other students.

6.2 When working off-campus, you will need a Windows-based PC with a three-button mouse and a strong, reliable Internet connection. MacBooks WILL NOT be supported for this course. You can view the minimum computing requirements here under Engineering Technology Majors.

6.3 This course cannot be completed using only a cell phone.

6.4 You must have access to and know how to use Purdue's Brightspace.

6.5 You may be required to take screen captures and add them to a PowerPoint document or another presentation application's document. Identify the app you will use to take screenshots and learn how to use it. Images from a cell phone are not sufficient.

6.6 You must know how to save a file from PowerPoint or other presentation applications as a PDF.

6.7 You must have access to and know how to use Adobe Acrobat (PDF Reader).

6.8 You may be required to know how to manipulate ZIP files and other basic computer file manipulations (7-Zip/WinRAR/WinZip).

6.9 You will need access to a scanner or have a scanner app on your phone.

6.10 You will need to have access to a printer.

6.11 You may need to make screen capture videos. Kaltura Capture, a screen capture video software, is available in Brightspace.

6.12 Mechanical pencil size 0.5 or 0.7 and eraser (plastic Staedtler, Pentel suggested).

6.13 Felt tip pen non-bleed (for example, Fine Point (0.4), black. DO NOT USE BALL POINT!

6.14 Required Software

6.14.1 You **do not** have to purchase or install any required software on your PC. All software needed is installed in the labs that are scheduled for this class or available via AppsAnywhere. The NX application provided in labs and through AppsAnywhere is configured to have a direct connection to the Product Data Management (PDM) system used in the course. A locally installed version of the software is not compatible and not supported.

6.14.2 Be aware that some of the software may **NOT** be available in other labs on campus, or they may have different versions installed. Check the version before using them to ensure it is the same or a newer version used in the class, or we may not be able to open your work. *(For example, files created in NX 2306 cannot be opened in NX 2027). This course will use Managed NX 2306 Integrated with Teamcenter 2312.*

6.14.3 If you do need to use the software outside of class, please check the lab schedule that is usually posted on the lab door and found on the course webpage: https://prof1st.square.site/lab-information.

6.14.4 If you need assistance with remote access, contact ECN. You can file a Trouble Report here: https://engineering.purdue.edu/ECN/AboutUs/ContactUs/pySubmit or call 765 494-4326

6.14.5 Should you choose to use the required software on your PC, please note that **access on a personal computer is a convenience** and usually might need a fair amount of troubleshooting to get it working correctly. Because the demos and procedures are run on Windows-based machines, Linux and MacBook users may need to contact ECN to configure and troubleshoot their personal computers.

7 Office Hours

7.1 The course instructor's office hours will be conducted by appointment. If you would like to meet, please email the instructor your available times, and a meeting will be scheduled. Meetings can also be conducted virtually using MS Teams.

7.2 TA office hours can be found at the top of this document or on the course website: OFFICE HOURS.

7.3 TA office hours allow students to discuss questions related to the material with a TA. Take advantage of office hours if you have questions about course material, not with the intent of completing the assignment during office hours.

8 Multiple Class Sections

8.1 This is a two-credit-hour course; it meets two times per week. You should be enrolled in two different sections with two different section numbers:

8.2 **Lecture** – **Schedule Type:** Lecture
Lab - Schedule Type: Laboratory

8.3 The lectures section is for Theory Lectures. Lab Preps are posted online to assist with the week's lab assignments. The lab sections are when you meet with your TA and work on weekly assignments. The sections are described in more detail in **Course Logistics** below.

9 Lab Section Number & TA Name

9.1 Memorize your **Lab Section** number and **TA Name**. Your **lab** section number is **NOT** the same as your lecture section number or your lab prep lecture section number. The lab section times are listed at the end of this document for your reference.

10 Email Instructions

10.1 All course-related emails for assistance should go to the MFET Help Form Submission, which creates a fully defined email for the quickest resolution possible.

10.2 The form sends an email to a Group Email box to allow TAs to work as a team to answer student questions and provide the quickest response to students possible.

10.3 If you need to contact the TA or Professor about a personal, non-course content-related issue, send it directly to the TA on the "TO:" line and the Professor on the "CC:" line and include the **Course Number in the Subject**.

10.4 Use your Purdue email (**NOT** Gmail or Comcast, etc.)

10.5 Do **NOT** expect emails to be answered between the hours of 7PM and 7AM Eastern.

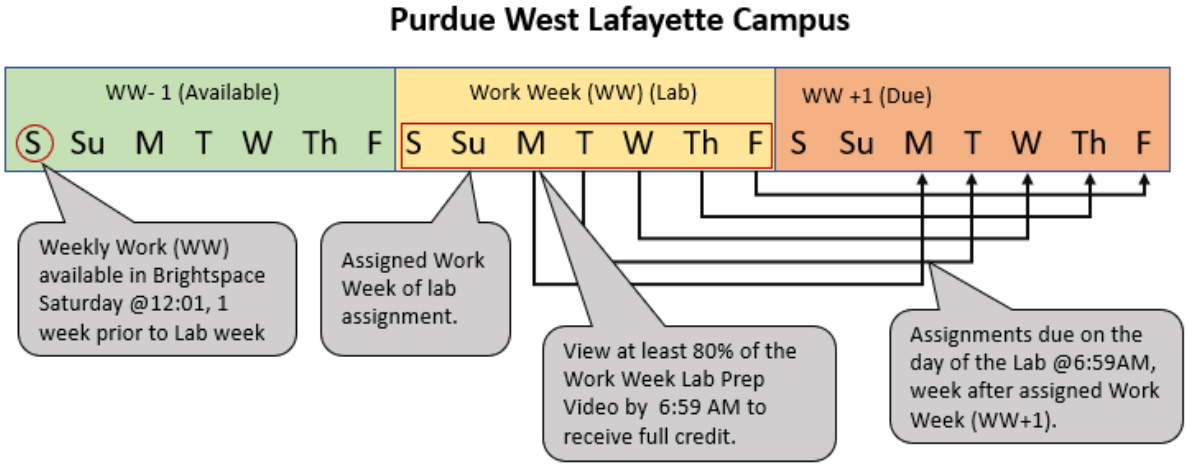
10.6 Emails will usually be answered within 24 hours or sooner, by a TA (weekends and holidays may have a longer window of response).

11 Course Schedule

11.1 Weekly breakdown of topics and due dates. All due dates in Brightspace reflect the day and time of assignment submission. If a due date is not present in Brightspace, the following schedule applies.

11.2 Topic schedule, Lab Assignment & Project submission schedule. (Link due date spreadsheet)

Work Week (WW)	Topic	Available 12:01 AM EST	Lab Assignment/Project Due Dates (These dates supersede any Brightspace dates)	
Week 1	Course Introduction/Hand Sketching	12-Jan	6:59 AM EST of	Week 2 Lab Day
Week 2 | MLK DAY	Parametric Sketching & Solid Modeling	12-Jan		Week 3 Lab Day
Week 3	CAD Systems & Part Modeling	17-Jan		Week 4 Lab Day
Week 4	Intoduciton to the Digital Enterprise	24-Jan		Week 5 Lab Day
Week 5	PDM	31-Jan		Week 6 Lab Day
Week 6	Parameters & Best Practices	7-Feb		Week 7 Lab Day
Week 7	**EXAM -1 / PRACTICAL 1** PDM Basics & Product Data	14-Feb		Week 8 Lab Day
Week 8	Assemblies & Products	21-Feb		Week 9 Lab Day
Week 9	Engineering Documentation	28-Feb		Week 10 Lab Day
	SPRING BREAK			
Week 10	Representing Processes	7-Mar		Week 11 Lab Day
Week 11	Representing Change	14-Mar		Week 12 Lab Day
Week 12	Organizations & Security in PDM	21-Mar		Week 13 Lab Day
Week 13	**EXAM 2 |** Guest Lecture	28-Mar		Week 15 Lab Day
Week 14	Guest Lecture	4-Apr		Week 16 Lab Day
Week 15	**Practical-2 |** Course Wrap-up	11-Apr		Week 16 Lab Day
NOTE: Quizzes, disucssions, survey's and other non-CAD or Sketching related assignments are due by prescribed date in Brigthspace.				



12 Course Logistics

12.1 The course material is arranged by weeks.

12.2 A typical week includes:

- 12.2.1 Lab Prep (Online Recording)
- 12.2.2 Content Lecture
- 12.2.3 Reading(s)
- 12.2.4 Chapter Quiz(zes)
- 12.2.5 Assignment(s)

12.3 **Lab Prep** is a prerecorded lecture that helps you prepare for the lab assignments coming up. These lectures focus on how you will use the CAD, PDM system, or other tools to accomplish your assignments. There are tips and additional information in the lab prep lecture that will be needed to complete your lab promptly, so attending lectures is a good idea. Even if you do not need guidance on your assignments, there will be information in the lab prep lecture that you will need to know and will be responsible for. Lab Prep demos **WILL BE** posted to Brightspace for reference in labs.

12.4 **Content Lectures** support the assignments or objectives. Content Lectures differ from Lab Preparation Lectures in that the information in content lectures applies to CAD systems in general. Material includes the role of CAD in product design, the history and scope of CAD, parametric solid modeling of parts and assemblies, and downstream uses of parametric solid modeling. Product Data Management (PDM) and its connection to CAD, Model-Based Definition (MBD), and technical drawings will also be covered. You are responsible for the material covered in the Content Lectures exams. Content lecture slides **WILL BE** posted to Brightspace.

12.5 **Readings** are derived from the required textbooks or articles that will support the assignments or objectives. You are responsible for the material covered in the readings on quizzes and exams.

12.6 **Quizzes** will cover the weekly textbook reading(s) as well as other content in the course. Textbook-related quizzes will be administered using the textbook publisher's website. Each chapter will have a separate quiz (5 questions each). Unlimited attempts are permitted; however, only the highest score of the first two attempts will be recorded in Brightspace.

12.7 **Assignments** (also called labs) will have a **Requirements** document. The Requirements document will specify anything required for the assignment. Assignments are typical exercises that will be in the CAD system, hand sketching, or similar. There will be written and/or video instructions for most assignments. Other assignments will be based on previous assignments; however, they will not have a step-by-step guide. As the course progresses, subsequent labs or assignments will build on previous assignment material. Therefore, the assignment will have fewer details as the semester progresses. You are expected to know how you did something in a prior assignment for a future assignment.

12.8 Lab Operations

- 12.8.1 To receive credit for your assignments, you must have your assignments submitted to the appropriate system (Brightspace or Teamcenter, depending on the assignment requirements) by the prescribed due date.
- 12.8.2 All work will be graded outside of the lab, and feedback will be provided through the appropriate medium (Brightspace or Teamcenter).
- 12.8.3 The lab instructor's primary role in the lab is to assist in the completion of the current week's assignments. It may not be possible for all students to complete lab assignments in the prescribed lab time. Take advantage of the time you have in the lab and office hours (for questions) if you do not have time to complete the assignment in the lab.
- 12.8.4 To build camaraderie within the lab and a network of assistance, when help is needed in the lab, first ask your peer to your left, right, front, or back before calling on the TA. Your fellow students might have already solved the issue, and this provides them the opportunity to teach, because "*you never learn so much as when you teach*".
- 12.8.5 If a student desires more detailed feedback on graded assignments, they can request it from the instructor and/or TA via the MFET Help Form.

12.8.6 Lab attendance is part of the overall score for the course, which ensures students effectively utilize the lab time set aside by the university and TA to complete weekly assignments. To receive credit for attending the lab, students must arrive within 15 minutes of the lab starting and may only sign out of the lab early if they have completed the work that is due for the next week. (See 14-Attendance & Participation and 16-Grading Scale)

12.8.7 Assignments will be checked to uphold academic integrity. It is strongly discouraged for students to share files with other students, regardless of reason (even for reference) or to conduct work on another student's personal computers or university logins (See 21-Academic Integrity). If it is determined that multiple User IDs were used to create a file, this will constitute a breach of academic integrity for all involved User IDs.

12.9 Exams

Two exams will be conducted using Brightspace Lockdown Browser. Each exam will consist of 30 questions from the lecture and textbook content, with a time limit of 30 minutes per exam. Questions from the textbook may not directly appear on the chapter quizzes. If the exams are not completed by the due date, makeup exams will not be permitted unless prior coordination has been made with the instructor.

12.10 Lab Practicals

Lab Practicals will be conducted through Brightspace, utilizing the software tools and techniques taught in the course. If you require extended time, any necessary accommodation will be provided in Brightspace. Lab Practicals exist to measure modeling competency. Students will be required to model parts with a specific set of requirements in a specified amount of time.

13 Deadlines & Submissions

13.1 **Deadlines** are an **unavoidable** part of being a professional. Act like a professional, and you will be treated like a professional.

13.2 All assignments will be assigned online on Brightspace.

13.3 Assignments will be submitted per the assignment requirement documents, which will specify either Brightspace or Teamcenter.

13.4 Each assignment must be in the prescribed format in the individual assignment requirement only. ALL other formats are UNACCEPTABLE.

13.5 Missed or late assignments will receive zero points unless the student has arranged in advance to complete the work due to a documented and excusable circumstance.

13.6 The two (2) lowest Assignment scores will be excluded from the final grade calculation. (This does not include Project Assignments)

13.7 The two (2) lowest Quiz scores will be excluded from the final grade calculation.

13.8 Extensions or accommodations will only be granted in cases of absences officially authorized by the Dean of Students, or upon receipt of formal documentation for participation in conferences, interviews, or other activities directly tied to academic or professional growth.
(See Section 14-Attendance & Participation Policy)

13.9 Students are expected to begin assignments in advance and allocate sufficient time for completion well before the deadline. Technical issues occurring near the submission deadline will not be accepted as a valid excuse. Promptly seek assistance from ITaP or ECN when technical problems arise.

13.10 Submissions or promotions in Teamcenter that fail due to server-related technical issues occurring after 4:30 PM on the day before the deadline will not be eligible for extension or credit. Since this falls outside regular IT support hours, students are strongly encouraged to complete and submit their work well in advance of the due date. (The policy will only vary from syllabus section **13.9** for server-related events.)

13.11 All assignments, projects, quizzes, practicals, and exam grade disputes must be made by the student within three (3) days of the grade and feedback being posted in Brightspace to be addressed by the TA or Instructor. After 3 days, all grades will be considered final.

13.12 Any assignment or project submitted early will be subject to early grading. If a request is made to change or update an item before the deadline, it will only be granted if the item has not already been graded.

13.13 When a student joins the course late within the first two weeks and the lab for the given week has already passed, and the student was unable to attend, the due date for the student's work for that week will automatically be extended to the due date for the following week.

14 Attendance & Participation Policy

14.1 It is expected that students will attend all lectures, as attendance will be taken via iClicker interactions and counted towards the final grade. Lecture participation will be recorded via **iClicker.** During lectures, students will be required to answer 2-3 questions at the end of each lecture on the material covered. The correctness of the 2-3 questions will dictate the other portion of the Lecture Attendance score. One Point will be granted for answering at least one question, and 2.5 points for a correct answer, for a total of 5-7.5 points for each lecture period. Incorrect answers will receive a score of 0. If the **iClicker** does not function correctly during the lecture, you must sign into the notebook at the end of the lecture to receive the participation score, and the instructor will grant one correct answer for that lecture. If a student signs in to a lecture using **iClicker** and is found not to be present, it will be recorded as a violation of academic integrity. The first offense will result in a warning. The second offence will be reported to the Dean of Students. The third office will be reported to the Dean of Students, and the student will fail the course.

14.2 Lab attendance is required, and attendance will be taken at the end of the lab via a sign-in sheet or when a student has correctly submitted all work for the current week for evaluation and has it verified by the TA.

14.2.1 **Students must attend their assigned lab and arrive within the first 15 minutes of the lab start to receive attendance credit.**

14.2.2 **Students are expected to watch ALL recorded lab prep videos with 80% completion by Monday at 6:59 AM EST of the current work week.** Starting with **Week 01 and Week 02** Lab Preps (due January 20th, 6:59 AM EST with a 1-day delay for MLK Day). Any completion rate below 80% will receive the corresponding percentage of that grade.

14.3 **Students may miss two (2) Lectures and two (2) Lab periods FOR ANY reason** such as medical, family emergencies, participation in university activities such as athletics, arriving late on campus at the start of the semester or after breaks, ROTC commitments, industry networking, interviews, etc., etc., etc. (For university sponsored events, a letter can be submitted to the MFET Help From to excuse the absence in addition to the three given absences.)

14.4 In the case of an absence being granted by the Office of the Dean of Students, it will not count against the allotted number of excused absences. There will not be any extensions for assignments, quizzes, or projects unless excused by the Dean of Students.

15 Evaluation (relative weighting)

Graded Item Categories	Pts./Percentage of Overall Course Grade
Assignments	25%
Project	25%
Exams/Lab Practicals	25%
Quizzes	15%
Lecture Attendance	5%
Lab Attendance	3%
Lab Prep Video	2%
Total	**100%**

16 Grading Scale

16.1 Letter grades will be based on the percentage of total points earned for the entire semester as follows:

Ranges		
A+ = 100 - 97%	**A** = 96.99% - 93%	**A-** = 92.99% - 90%
B+ = 89.99% - 87%	**B** = 86.99% - 83%	**B-** = 82.99% - 80%
C+ = 79.99% - 77%	**C** = 76.99% - 73%	**C-** = 72.99% - 70%
D+ = 69.99% - 67%	**D** = 66.99% - 63%	**D-** = 62.99% - 60%
F = 59.99% or below		

17 Successful Participation Guidelines

17.1 If you want to be a successful student in this course:

17.1.1 Be self-motivated and self-disciplined.

17.1.2 Let us know if problems arise when they arise. Don't wait to contact the instructor team.

17.1.3 Be willing and able to commit to doing a week's worth of work in a day.

17.1.4 Be able to communicate through writing at a college level.

17.1.5 Be able to meet the minimum requirements for the course.

17.1.6 Use critical thinking and decision making as part of the learning process.

17.1.7 Acknowledging the following directions is an important and serious part of this course.

17.1.8 Stay organized and up to date.

17.1.9 Plan for unforeseen technical difficulties in the submission of your assignments.

17.2 In contrast, here are some common behaviors that lead to failing the course…

17.2.1 Wait until the last minute to start and/or submit assignments.

17.2.2 Forget about deadlines.

17.2.3 Ignore emails from the instructor and/or your peers regarding course activities.

17.2.4 Don't get familiar with the grade book and syllabus.

18 Lab Section/Lab Times/Lab Location/Lab TAs

19 Use of Copyrighted Materials

19.1 Students are expected, within the context of the Regulations Governing Student Conduct and other applicable University policies, to act responsibly and ethically by applying the appropriate exception under the Copyright Act to the use of copyrighted works in their activities and studies. The University does not assume legal responsibility for violations of copyright law by students who are not employees of the University.

19.2 Copyrightable Work created by any person subject to this policy is primarily to express and preserve scholarship as evidence of academic advancement or academic accomplishment. Such works may include, but are not limited to, scholarly publications, journal articles, research bulletins, monographs, books, plays, poems, musical compositions, and other works of artistic imagination, and works of students created in the course of their education, such as exams, projects, theses or dissertations, papers, and articles. Please read the University Regulations on policies:

19.3 (http://www.purdue.edu/policies/academic-research-affairs/ia3.html).

19.4 Any student found publishing or sharing copyrighted materials illegally to any site or venue (Reddit, Google Drive, OneDrive, Quizlet, etc.) will be reported to the Dean of Students, the publisher, and the copyright owner of the material for potential litigation under the Digital Millennium Copyright Act (DMCA).

20 Commercial Note Taking

20.1 The information in this section is taken from Part 10 of the Purdue student miscellaneous conduct regulations (University Senate Document 03-9, April 19, 2004), available at http://catalog.purdue.edu/content.php?catoid=8&navoid=8208#miscellaneous-conduct-regulations

20.2 As used in this paragraph, the term “instructor” is defined as the individual who authored the material being presented as part of the course.

20.3 Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

20.4 Notes taken in class are, however, generally considered to be “derivative works” of the instructor’s presentations and materials, and they are thus subject to the instructor’s copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor. To obtain permission to sell or barter notes, the individual wishing to sell or barter the notes must be registered in the course or must be an approved visitor to the class. Course instructors may choose to grant or not grant such permission at their discretion and may require a review of the notes before they’re being sold or bartered. If they do grant such permission, they may revoke it at any time, if they choose.

21 Academic Integrity

21.1 Working Collaboratively - Working together is encouraged, but completing assignments for other students or sharing files is not allowed. This includes doing a "File, Save As" and providing another student with the file. This activity can be traced through file history and will result in immediate failure of the class and reporting of all involved students to the Office of the Dean of Students.

21.2 Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information submitted, the greater the opportunity for the University to investigate the concern. More details are available in Brightspace, under University Policies.

Any student **submitting work** that is **not their own** will be **reported** to the **Dean of Students**.

Any student's work that is submitted by another student in this class or future classes will also be reported to the **Dean of Students** and considered for **discipline in the course,** up to the **potential of failure of the course** if the instructor determines that **academic integrity has been breached**.

Be advised that using another student's laptop or computer account will flag a file as being owned by the owner of the laptop and/or user account.

Any **student assisting** (including the posting of course materials such as quiz questions, exam questions, or CAD models online) is engaging in academic dishonesty.
and will **be reported** to the **Dean of Students** and considered for **discipline in the course** up to the **potential of failure of the course** if the instructor determines that **academic integrity has been breached**.

Students enrolled in this class will **immediately fail the course if academic integrity is breached.**

(Example: submission of another's work as their own)

Two offenses can get you expelled from the University.

21.3 Submitting work that is not your own includes: participating in **cheating, plagiarism, copying material** from another person's media, using illegal cribs online or other materials during a written examination, lying to course instructors and lab assistants about their own work, stealing tests, quizzes, or answer keys, and any such activities will be considered in conflict with the printed academic honesty guidelines as set out by Purdue University and the Purdue Polytechnic Institute. In such cases, the matter will be reported to the Office of the Dean and the appropriate Purdue University officers for consideration and possible disciplinary action.

21.4 Use of Copyrighted Materials: Materials presented in this class are protected by copyright law and must be cited if used for reference. Students enrolled in and authorized visitors to Purdue University courses are permitted to take notes, which they may use for individual/group study or other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

21.5 Standards set by Purdue University, as outlined in the Student Handbook and the University Regulations, will be observed in this course. Students with special needs, including those who are hearing or visually challenged, or in need of tutoring, may contact the Dean of Students' Office located in Schleman Hall, Room 207, or call 765-494-1747 for assistance.

21.6 The use of AI resources to complete graded work, such as discussion questions, is not permitted in this course. If an answer is determined to have been generated by AI, the submission will receive a zero, and the student may be reported to the Dean of Students for an Academic Integrity violation.

22 Netiquette

22.1 Your instructors and fellow students wish to foster a safe online learning environment. All opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse. You are encouraged to comment, question, or critique an idea, but you are not to attack an individual. Please consider that sarcasm and humor can be misconstrued in online interactions and generate unintended disruptions. Working as a community of learners, we can build a polite and respectful course ambiance. Please read the Netiquette rules for this course:

22.2 Do not dominate any discussion. Allow other students to join in the discussion.

22.3 Do not use offensive language. Present ideas appropriately.

22.4 Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.

22.5 Avoid using vernacular and/or slang language. This could lead to misinterpretation.

22.6 Keep an “open mind” and be willing to express even your minority opinion.

22.7 Think and edit before you push the “Send” button.

22.8 Do not hesitate to ask for feedback.

23 Mental Health/Wellness

23.1 If you find yourself beginning to feel some stress, anxiety, and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression, and other concerns. It provides accessible and effective resources, including short videos, brief exercises, and self-reflection tools.

23.2 If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are Monday through Friday, 8 a.m. to 5 p.m.

23.3 If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

23.4 If you’re struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

24 Emergency Preparedness

24.1 In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted on the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email frequently.

24.2 A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the Emergency Preparedness for Classrooms document.

24.3 The first day of class, I will review the Emergency Preparedness plan for our specific classroom, following Purdue's required Emergency Preparedness Briefing. Please make note of items like:

24.3.1 The location to which we will proceed after evacuating the building if we hear a fire alarm.

24.3.2 The location of our Shelter in Place in the event of a tornado warning.

24.3.3 The location of our Shelter in Place in the event of an active threat, such as a shooting.

25 Student Help and Success Resources

25.1 Purdue offers so many opportunities to get help if you are struggling. Check Purdue's website, purdue.edu, for the latest information on these services. Links to many policies (accurate at the time of publication) are in the Student Services and Resources module of the course content on Brightspace. These include the Academic Success Center, the world-renowned Purdue Online Writing Lab (OWL), Counseling and Psychological Services (CAPS), and others.

26 University Policies

26.1 The latest version of the Purdue University policies is found on the Purdue website, purdue.edu. Links to many policies (accurate at the time of publication) are in the University Policies module of the course content on Brightspace. These include Emergency Preparedness, Grief Absence Policy, Accessibility Policies, Guide to Academic Integrity, and others.

27 End of Course Communication

SUBJECT: MFET End of Course

Hello MFET,

Final grades have been posted.

We would like to take this opportunity to thank all students enrolled in MFET this semester, for the professional and courteous manner that was displayed, and afforded to me and your TAs.

We thoroughly enjoyed teaching this course, and I hope you have gained some insight into the importance of CAD, PLM, and the Digital Thread in professional environments, that will hopefully assist you in your chosen career. As the we finish the grades, and everyone knows their respective grade for the semester, some students are a few points away, (sometimes only 0.1 point away) from their desired grade. Unfortunately, I cannot add points or open the assignments that are closed to allow these students to attempt to achieve additional points.

This situation occurs every academic semester, and it is completely unfair to the students that did the work and turned in the assignments on time, and it devalues their earned grade. The idea of adding points would create a slippery slope that cannot be allowed in this course to maintain academic integrity.

We empathize with each student who did not receive the grade that they desired, and I consistently tried throughout the semester to offer opportunities for counseling, help sessions, answering questions, and responding to emails in a prompt manner. I know that the students affected will be disappointed with this response, but I do feel that it is the only fair academically consistent approach to take.

If you have any questions or further concerns, please feel free to contact me, as I would be willing to talk to you individually, (this does not mean I will adjust your grade, however).

Thank you, and have a wonderful, relaxing, and great break!

Best regards,

Prof. Fuerst

28 Disclaimer

28.1 This syllabus is subject to change. Significant changes will be announced in Brightspace.